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| Animals | Most aquatic animals | Birds and most reptiles, insects and land snails | Mammals, most amphibians, sharks, reptiles and terrestrial invertebrates |
| Nitrogenous wastes | Ammonia | Uric acid | Urea |

**Figure 8.1** Excretory products in different groups of animals.

uric acid are produced to conserve water. Mammals and terrestrial amphibians mainly excrete urea and are called **ureoteles**. Earthworms while in soil are ureoteles and when in water are ammonoteles. Figure 8.1 shows the excretory products in different groups of animals.

The animal kingdom presents a wide variety of excretory structures. Most invertebrates have a simple tubular structure in the form of primitive kidneys called **protonephridia** and **metanephridia.** Vertebrates have complex tubular organs called kidneys. Protonephridia are excretory structures with specialized cells in the form of **flame cells** (cilia) in Platyhelminthes (example tapeworm) and **Solenocytes** (flagella) in *Amphioxus*. Nematodes have **rennette cells, Metanephridia** are the tubular excretory structures in annelids and molluscs. **Malpighian tubules** are the excretory structures in most insects. **Antennal glands** or **green glands** perform excretory function in crustaceans like prawns. Vertebrate kidney differs among taxa in relation to the environmental conditions. Nephron is the structural and functional unit of kidneys. Reptiles have reduced glomerulus or lack glomerulus and Henle's loop and hence produce very little hypotonic urine, whereas mammalian kidneys produce concentrated (hyperosmotic) urine due to the presence of long Henle's loop. The Loop of Henle of the nephron has evolved to form hypertonic urine. Aglomerular kidneys of marine fishes produce little urine that is isoosmotic to the body fluid. Amphibians and fresh water fish lack Henle's loop hence produce dilute urine (hypoosmotic).

The average bladder holds between 300ml and 600ml of urine. If the urinary system is healthy, urine may stay in the bladder for up to about 5 hours before excretion, depending on the amount of liquid consumed. Nerves send signals to the brain when the bladder needs to be emptied, with this indication one will feel the urge to empty the bladder. The muscle in the bladder wall is called the 'detrusor' muscle. One may suffer from stress if the muscles supporting the bladder are weakened. Pelvic floor exercise helps to strengthen these muscles.

## 8.2 Human excretory system

### 8.2.1 Structure of kidney

Excretory system in human consists of a pair of kidneys, a pair of ureters, urinary bladder and urethra (Figure. 8.2). Kidneys



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